

EDUCATION

Missouri State University <i>M.Sc. (Ongoing) in Computer Science</i>	Springfield, MO <i>2025 – Present</i>
Khulna University of Engineering & Technology (KUET) <i>B.Sc. in Energy Science and Engineering, CGPA: 3.17/4.00</i>	Khulna, Bangladesh <i>2019 – 2024</i>
◦ CGPA of Last 60 Credits: 3.42/4.00 ◦ Notable Courses: Energy Storage Systems, Energy Efficiency Assessment, Power System Engineering	

RESEARCH EXPERIENCE

Missouri State University <i>Research Assistant, Department of Computer Science</i>	Springfield, MO <i>2025 – Present</i>
Khulna University of Engineering & Technology <i>Undergraduate Researcher</i>	Khulna, Bangladesh <i>2022 – 2024</i>
◦ Conducted numerical investigation of thermal performance of buildings integrated with Phase Change Materials (PCMs) in roof applications. ◦ Analyzed thermal functionality of PCMs in different climatic conditions across Bangladesh using DesignBuilder and EnergyPlus software. ◦ Investigated energy consumption reduction effects and evaluated percentage improvements in building energy efficiency. ◦ Developed automated single-axis solar tracking system for battery charging applications using Arduino and C++.	

PROJECTS

- **Numerical Investigation of Thermal Performance of Building Integrated with Phase Change Material:** Senior year thesis project utilizing DesignBuilder and EnergyPlus for comprehensive numerical investigation. Analyzed thermal functionality of PCMs integrated into single-story building roofs across different climatic conditions in Bangladesh. Evaluated energy consumption reduction percentages and thermal performance improvements.
- **Automated Single Axis Solar Tracking Based Battery Charging System:** Junior year project implementing solar tracking technology with Arduino microcontroller and C++ programming. Evaluated performance comparison between static and mobile solar systems. Designed and implemented efficient battery storage system for optimal energy utilization.

PUBLICATIONS

- **Conference Paper (Accepted):** Medur Rodoshi Aungshu, Dr. Mohammad Ariful Islam, "Numerical Investigation on the Thermal Performance of a Building Integrated with Phase Change Material in the Roof" International Conference on Mechanical, Industrial and Materials Engineering (ICMIME 2024), December 11-13, 2024, Rajshahi, Bangladesh, ID:403.
- **Journal Paper (Under Review):** Haque AF, Hossain MJ, Hossain MM, Shovon MAH, Aungshu MR, Galib MR, Alam MM. "Numerical Investigation on the Selection of Energy-Efficient Phase Change Material (PCM) for Building Envelopes in Different Climatic Conditions of Bangladesh." Energy and Buildings Journal.

HONORS & AWARDS

- **Merit Scholarships:** Received Merit Scholarships for a total of three semesters in the Department of Energy Science and Engineering at KUET, Bangladesh.
- **IELTS Academic Achievement:** Band Score: 7.0 (Listening: 8.0, Reading: 8.5, Writing: 6.0, Speaking: 6.0) - British Council, TRF Number: 24BD024370AUNM001A.

SKILLS

Programming Languages: C++, Python, Arduino Programming

Research Field: Energy Storage Solutions, Machine Learning for Materials Innovation, Energy Efficiency Improvements, Phase Change Materials, Solar Energy Systems